ICE 2345 DIGITAL COMMUNICATION

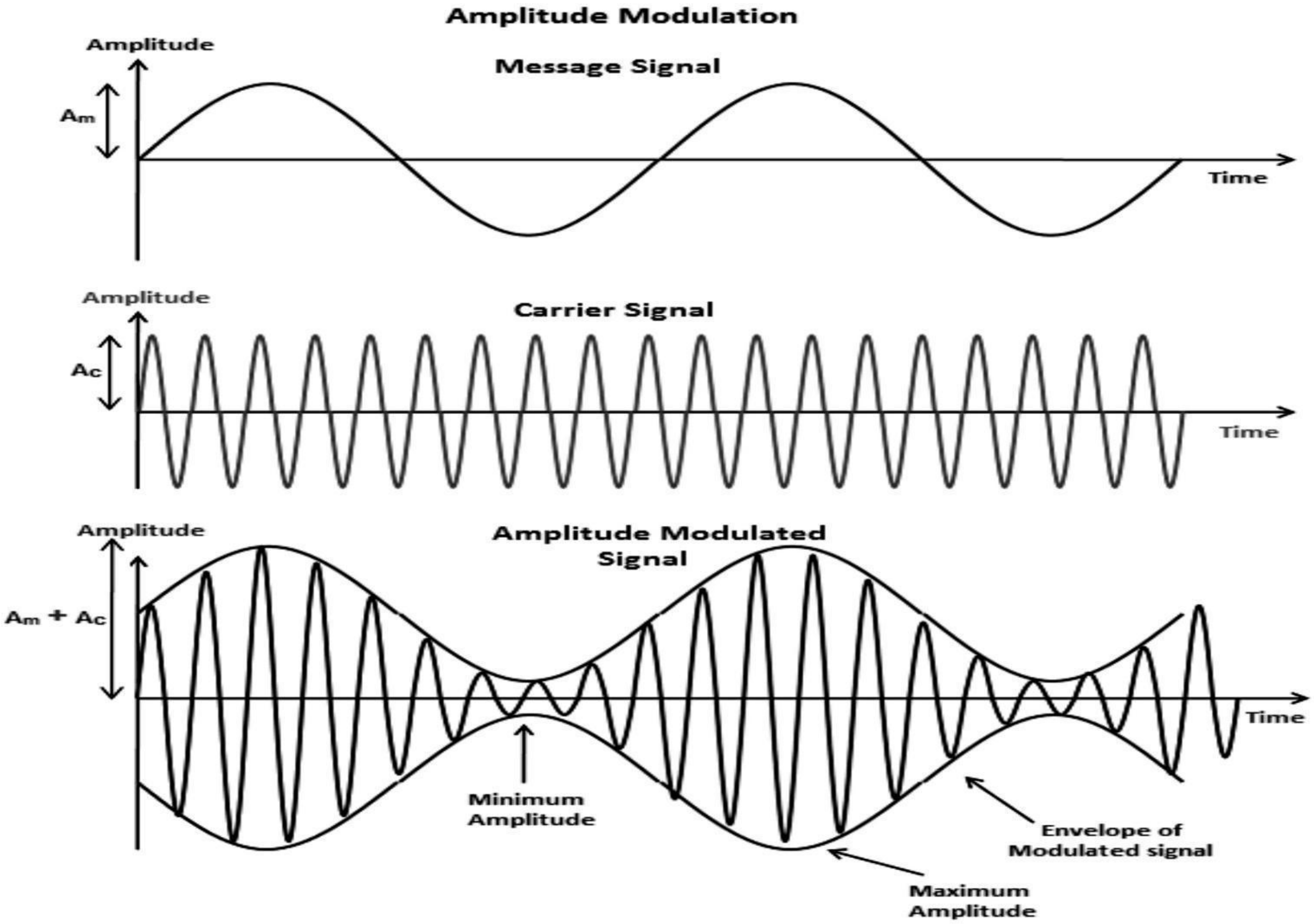
Abhirup Sarker
Lecturer
Department of Robotics and Mechatronics Engineering
Daffodils International University

AMPLITUDE MODULATION

What is amplitude modulation?
Amplitude modulation is the process of varying the amplitude of a high-frequency carrier signal in accordance with a low-frequency message signal. The frequency and phase doesn’t change of the carrier.

Why do we need amplitude modulation?
Low frequency signals can not travel long distance. Meanwhile high frequency signals can travel far, radiate easily and enable use of small antennas. It is used in broadcasting, distress signal, aviation communication etc.

AMPLITUDE MODULATION



AMPLITUDE MODULATION

Message/Modulating signal,

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal,

$$c(t) = A_c \cos(2\pi f_c t)$$

By principle of amplitude modulation,

$$s(t) = [A_c + m(t)]\cos(2\pi f_c t)$$
$$\rightarrow s(t) = [A_c + A_m \cos(2\pi f_m t)]\cos(2\pi f_c t)$$
$$\rightarrow s(t) = A_c[1 + \frac{A_m}{A_c}\cos(2\pi f_m t)]\cos(2\pi f_c t)$$
$$\rightarrow s(t) = A_c \cos(2\pi f_c t) + A_c \mu \cos(2\pi f_m t)\cos(2\pi f_c t)$$
$$\rightarrow s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2}\cos 2\pi(f_m + f_c)\, t + \frac{A_c \mu}{2}\cos 2\pi(f_m - f_c)\, t$$

Carier component

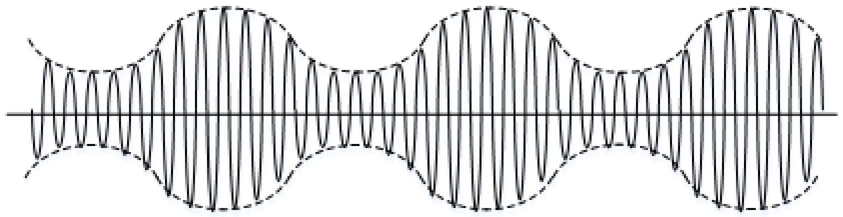
Upper side band

Lower side band

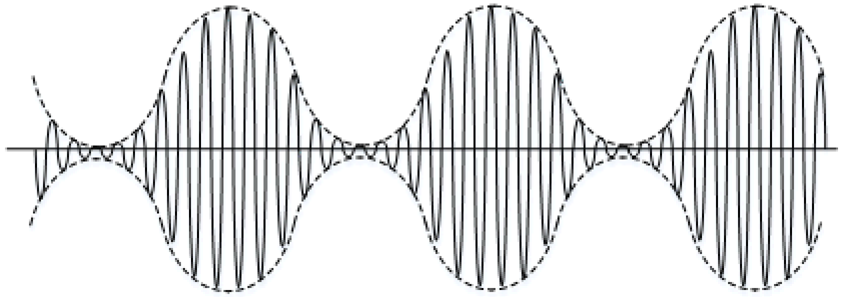
AMPLITUDE MODULATION

Modulating index, $\mu = \frac{A_m}{A_c}$

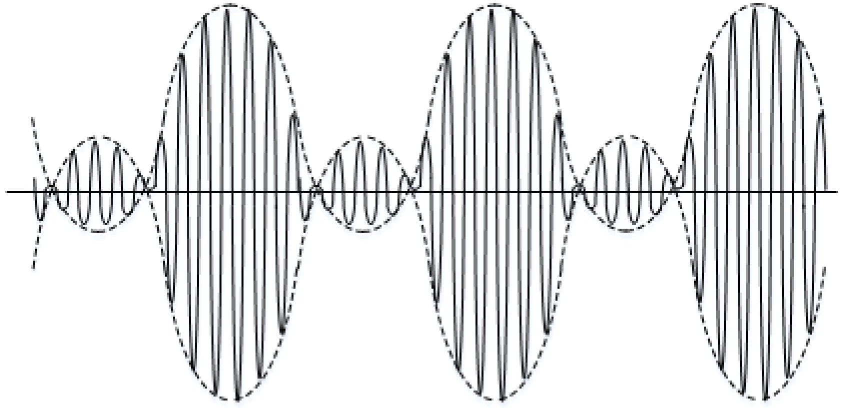
Under modulation μ<1



100% modulation μ=1



Over modulation μ>1



AMPLITUDE MODULATION

Total power transmitted , $P = P_c + P_{USB} + P_{LSB}$

Power of carrier,$P_c = \frac{A_c^2}{2}$

Similarly, $P_{USB} = \frac{1}{2}\frac{A_c^2\mu^2}{4}$ & $P_{LSB} = \frac{1}{2}\frac{A_c^2\mu^2}{4}$

So, sideband power $P_s = \frac{A_c^2\mu^2}{4}$

Total power transmitted , $P = P_c + P_s = P_c\left(1 + \frac{\mu^2}{2}\right)$

Efficiency, $\varepsilon = \frac{P_s}{P} = \frac{\mu^2}{2+\mu^2}$

DSBSC AMPLITUDE MODULATION

Double Sideband Suppressed Carrier or DSBSC is AM signal without carrier signal. We only send USB and LSB not the carrier signal for efficiency.

$m(t) = A_m \cos(2\pi f_m t)$ → Product Modulator → $y(t) = m(t)c(t)$

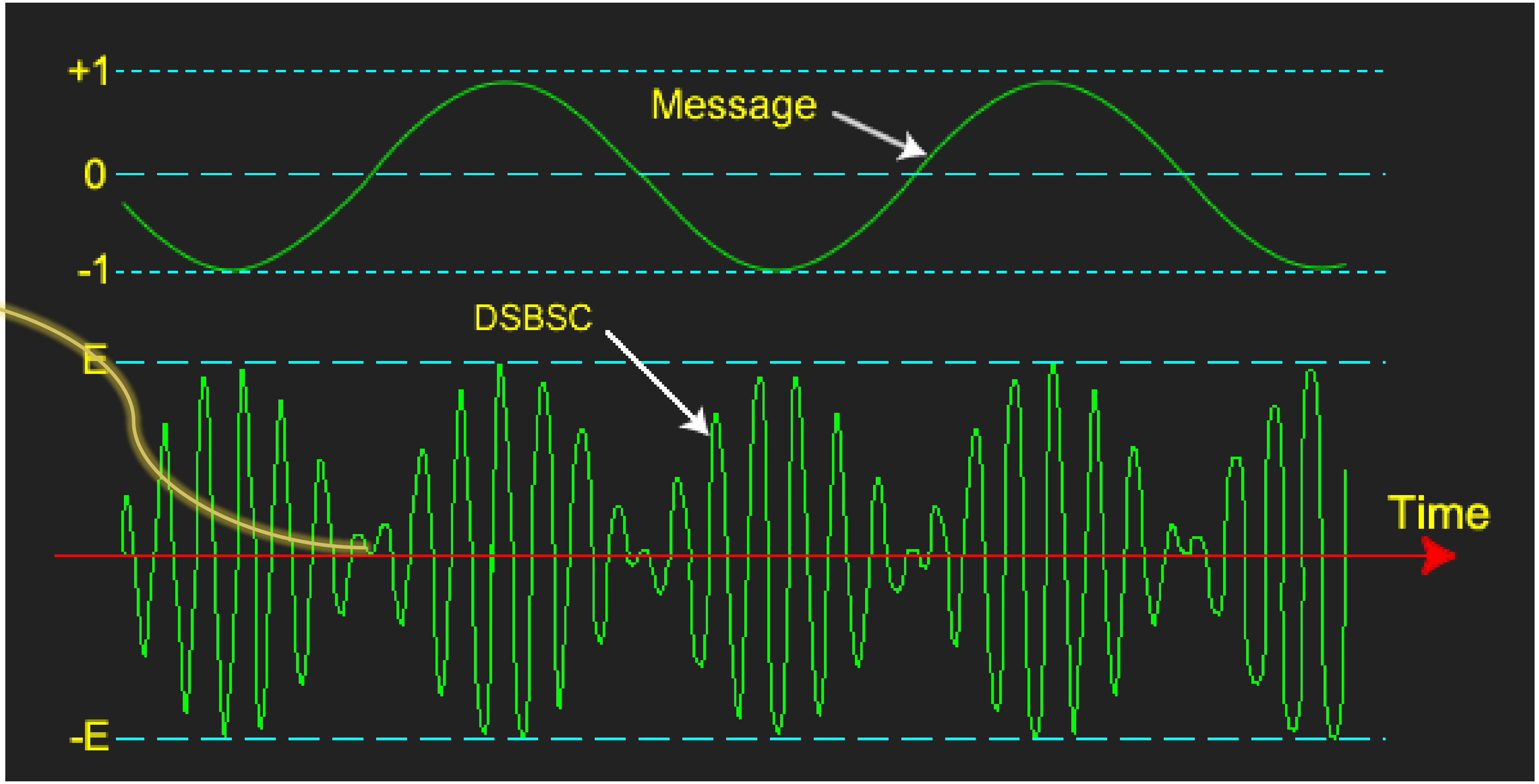
$c(t) = A_c \cos(2\pi f_c t)$

Characteristics:

- No carrier signal
- Sidebands carrying information
- 180 phase reversal at zero crossing of modulating signal
- Same bandwidth as AM
- Complex demodulation

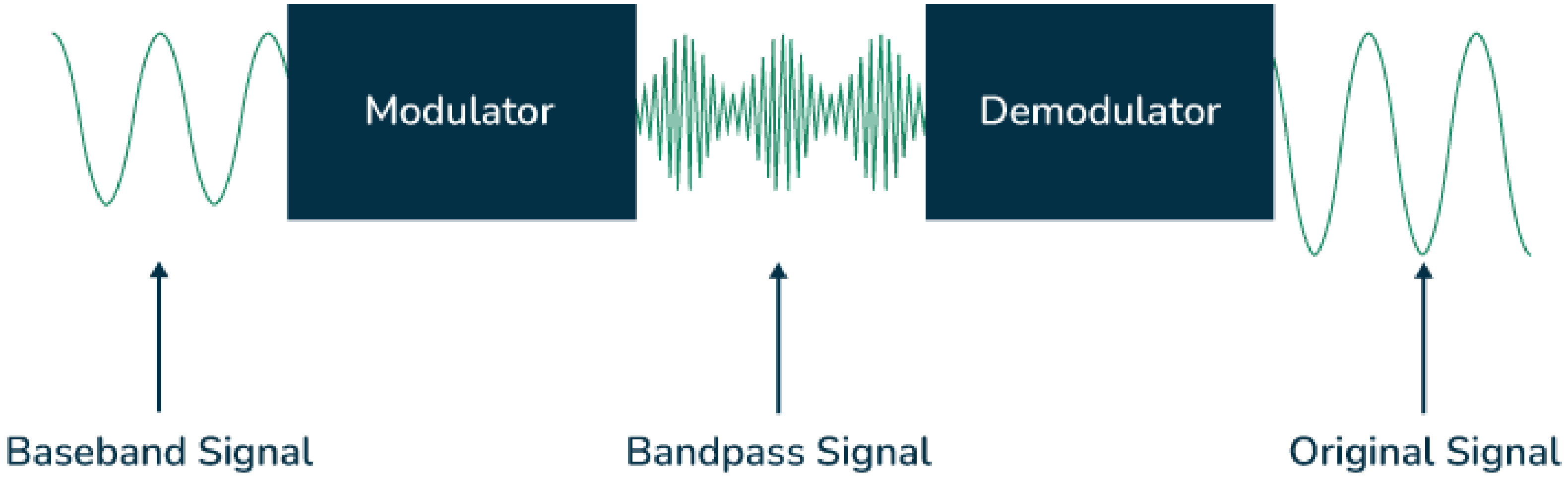
DSBSC AMPLITUDE MODULATION

Phase reversed



DEMODULATION

Demodulation is the process of extracting the original information or signal from a modulated signal. So, a demodulator will convert the carrier variation of amplitude, frequency or phase back to the message signal.



AMPLITUDE DEMODULATION

Types of amplitude demodulation:

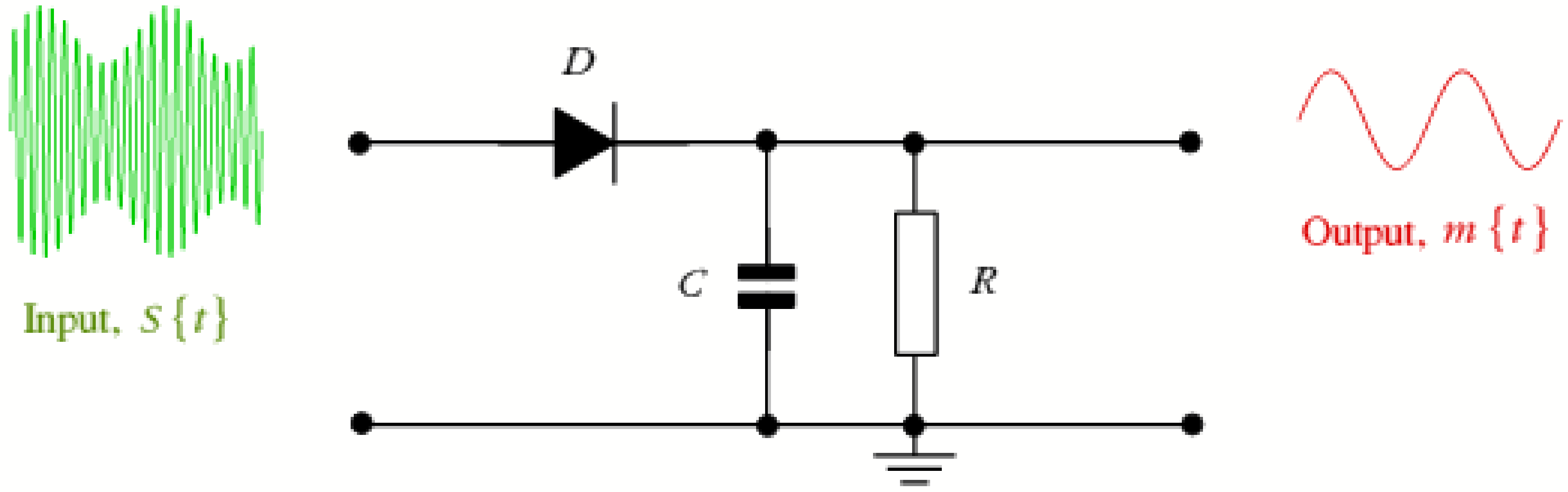
- ❑ Envelope Detector: uses diode RC circuit, works when carrier transmitted and signal is under or perfectly modulated. Its is very simple and cost-effective technique but not so efficient.
- ❑ Square Law: uses low pass filter to extract message, usually used for experiments which means very few practical implementation.
- ❑ Coherent/Synchronous Demodulation: can demodulate suppressed carrier signals, high efficiency and noise immunity but complex circuitry.

Importance of demodulation:

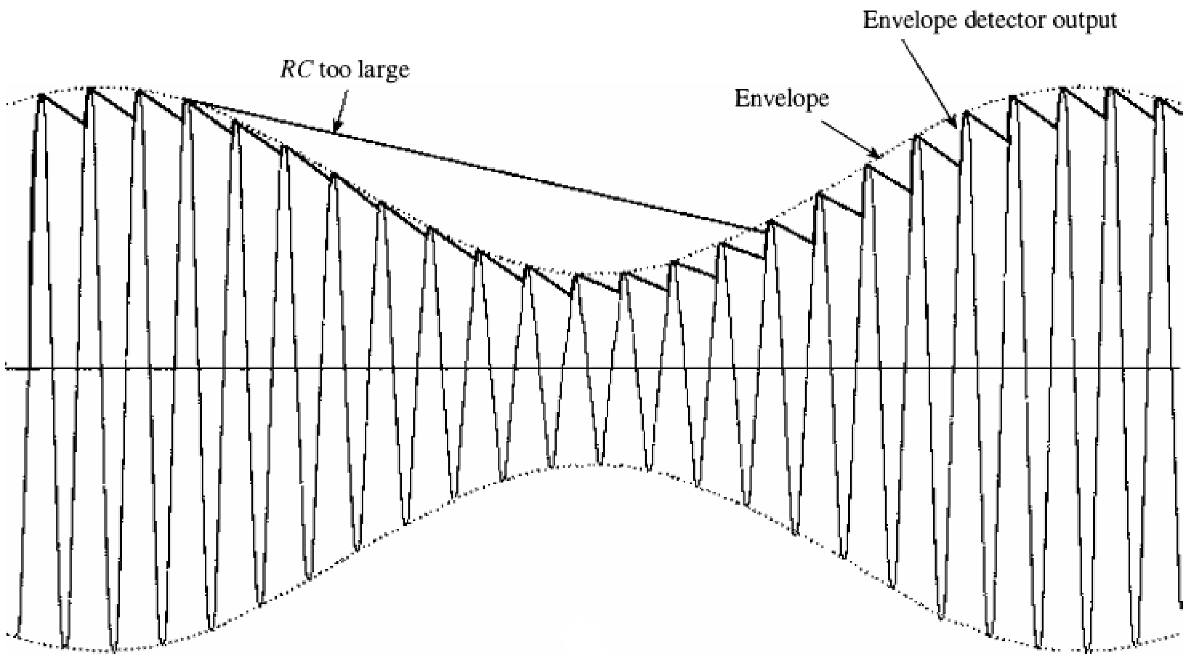
- Communication between transmitter and receiver.
- Recover original information.
- Provide noise immunity
- Application in radio-TV broadcasting, satellite communication, internet data transmission, radar and navigation etc.

ENVELOPE DETECTOR CIRCUIT

- ❖ Works when diode forward biased
- ❖ When voltage increases capacitor stores energy, then discharge when voltage drops
- ❖ Negative cycle is clipped
- ❖ Information loss

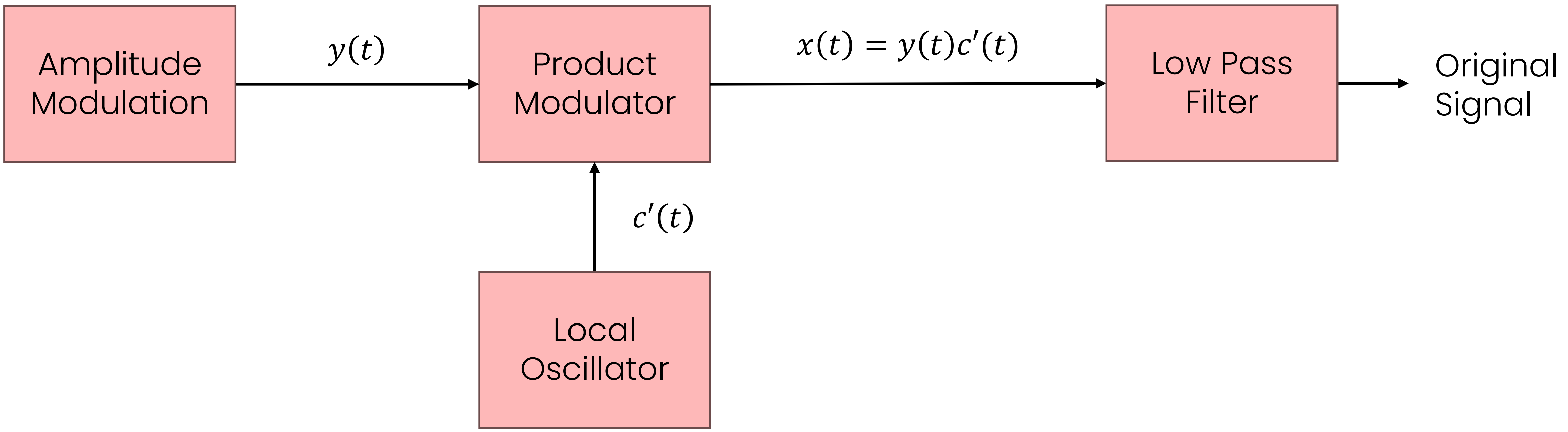


ENVELOPE DETECTOR CIRCUIT



- If RC is too small then capacitor discharges rapidly leading to more ripple
- On the other hand, if RC is too large it takes longer time to discharge and might not be able to follow the envelope.
- $\frac{1}{f_c} \ll RC \ll \frac{1}{f_m}$, for optimum performance.

COHERENT DETECTION



- Also known as synchronous detection
- Local oscillator generates carrier at receiver end at same frequency and phase with transmitter. This is important because carrier is not transmitted.
- High efficiency, noise immunity and useful for DSBSC.

LIMITATIONS

❑ Limitations of amplitude modulation:

- Large power waste in carrier
- Easily corrupted by noise
- Overmodulation is not suitable
- Limited bandwidth
- Bulky and expensive transmitter requirement

❑ Limitations of amplitude demodulation:

- Coherent detection is sensitive to phase synchronization and it has complex circuitry
- Envelope detector is noise affected easily, can’t work with over modulated signal.

THANK YOU